

AnalogInput

Generated by Doxygen 1.16.0

1 Directory Hierarchy	1
1.1 Directories	1
2 File Index	3
2.1 File List	3
3 Directory Documentation	5
3.1 AnalogInput Directory Reference	5
4 File Documentation	7
4.1 AnalogInput/AnalogInput.ino File Reference	7
4.1.1 Function Documentation	8
4.1.1.1 loop()	8
4.1.1.2 setRGB()	8
4.1.1.3 setup()	8
4.1.2 Variable Documentation	8
4.1.2.1 MAX_INPUT_VOLT	8
4.1.2.2 pinB	9
4.1.2.3 pinG	9
4.1.2.4 pinR	9
4.1.2.5 potPin	9
4.1.2.6 VREF	9
Index	11

Chapter 1

Directory Hierarchy

1.1 Directories

AnalogInput	5
AnalogInput.ino	7

Chapter 2

File Index

2.1 File List

Here is a list of all files with brief descriptions:

AnalogInput/ AnalogInput.ino	7
--	---

Chapter 3

Directory Documentation

3.1 AnalogInput Directory Reference

Files

- file [AnalogInput.ino](#)

Chapter 4

File Documentation

4.1 AnalogInput/AnalogInput.ino File Reference

```
#include <Arduino.h>
```

Functions

- void `setRGB` (int r, int g, int b)
Write PWM values to the RGB LED.
- void `setup` ()
Arduino setup function.
- void `loop` ()
Arduino main loop.

Variables

- const uint8_t `potPin` = A0
Analog input pin connected to the potentiometer wiper.
- const float `VREF` = 5.0
ADC reference voltage in volts.
- const float `MAX_INPUT_VOLT` = 5.0
Maximum expected input voltage from the potentiometer in volts.

RGB PWM pins

PWM output pins used to drive the RGB LED channels.

- const uint8_t `pinR` = 9
PWM pin for Red channel.
- const uint8_t `pinG` = 10
PWM pin for Green channel.
- const uint8_t `pinB` = 11
PWM pin for Blue channel.

4.1.1 Function Documentation

4.1.1.1 loop()

```
void loop ()
```

Arduino main loop.

1) Reads the ADC value from the potentiometer (0..1023). 2) Converts it into volts using VREF. 3) Prints the ADC value and voltage to Serial with 3 decimals. 4) Normalizes the voltage from 0..5.0V into a factor t in range 0..1. 5) Fades the RGB LED from purple to red by decreasing the blue channel.

4.1.1.2 setRGB()

```
void setRGB (
    int r,
    int g,
    int b)
```

Write PWM values to the RGB LED.

The input values are constrained to the valid PWM range (0..255) before being written with analogWrite(). For a common-cathode RGB LED, higher values mean brighter light for that color channel.

Parameters

<i>r</i>	Red channel intensity (0..255)
<i>g</i>	Green channel intensity (0..255)
<i>b</i>	Blue channel intensity (0..255)

4.1.1.3 setup()

```
void setup ()
```

Arduino setup function.

Initializes serial communication and configures RGB pins as outputs. Also turns the RGB LED off initially.

4.1.2 Variable Documentation

4.1.2.1 MAX_INPUT_VOLT

```
const float MAX_INPUT_VOLT = 5.0
```

Maximum expected input voltage from the potentiometer in volts.

In this exercise the potentiometer is wired between 5.0V and GND, so the wiper voltage range is 0..5.0V.

4.1.2.2 pinB

```
const uint8_t pinB = 11
```

PWM pin for Blue channel.

4.1.2.3 pinG

```
const uint8_t pinG = 10
```

PWM pin for Green channel.

4.1.2.4 pinR

```
const uint8_t pinR = 9
```

PWM pin for Red channel.

4.1.2.5 potPin

```
const uint8_t potPin = A0
```

Analog input pin connected to the potentiometer wiper.

4.1.2.6 VREF

```
const float VREF = 5.0
```

ADC reference voltage in volts.

Default Uno analog reference is ~5V (can vary slightly depending on USB power).

Index

AnalogInput Directory Reference, [5](#)

AnalogInput.ino

loop, [8](#)

MAX_INPUT_VOLT, [8](#)

pinB, [8](#)

pinG, [9](#)

pinR, [9](#)

potPin, [9](#)

setRGB, [8](#)

setup, [8](#)

VREF, [9](#)

AnalogInput/AnalogInput.ino, [7](#)

loop

AnalogInput.ino, [8](#)

MAX_INPUT_VOLT

AnalogInput.ino, [8](#)

pinB

AnalogInput.ino, [8](#)

pinG

AnalogInput.ino, [9](#)

pinR

AnalogInput.ino, [9](#)

potPin

AnalogInput.ino, [9](#)

setRGB

AnalogInput.ino, [8](#)

setup

AnalogInput.ino, [8](#)

VREF

AnalogInput.ino, [9](#)